GYANARANJAN ROUT

Semiconductor Physical Designer | RTL Design & Verification

Floorplanning • Placement & Routing • FPGA

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PROJECTS / EXPERIENCE

RTL-to-GDSII Physical Design of 32-bit RISC-V ALU

SkyWater 130nm | OpenLane, Yosys, OpenROAD, Magic, KLayout

- Designed and physically implemented a modular 32-bit RISC-V execution unit supporting RV32I and RV32M extensions, integrating a 32-bit carry-lookahead adder, barrel shifter, and iterative hardware divider.
- Executed the complete automated ASIC flow: synthesis, floorplanning, placement, CTS, and detailed routing using OpenLane on WSL2/Docker.
- Achieved timing closure with 0 setup violations and 0 hold violations at a 25.0 ns (40 MHz) clock period.
- Completed physical verification with 0 Magic DRC violations; streamed out a verified ~495 µm × 495 µm GDSII layout inspected in KLayout.

8-Bit ALU RTL Design and Verification

SystemVerilog | Xilinx Vivado, Yosys

- Implemented an 8-bit ALU supporting multiple arithmetic and logic operations.
- Developed a SystemVerilog verification testbench using directed and randomized vectors to validate corner cases.

Traffic Light Controller (FSM)

Verilog HDL | Xilinx Vivado

- Implemented an FSM controller for multi-way intersection signal transitions.
- Validated state-transition coverage through behavioral simulation and generated synthesized RTL schematics.

FPGA-Based Image Processing (RGB to Grayscale)

Verilog HDL | Xilinx Vivado

- Designed fixed-point logic to convert 24-bit RGB pixel streams into 8-bit grayscale intensity values.
- Verified pixel transformations and pipelined data flow using behavioral simulation.

EDUCATION

B.Tech in Electronics and Communication Engineering

Gandhi Institute of Excellent Technocrats, Bhubaneswar, Odisha
08/2024 – 05/2028

Higher Secondary Certificate (12th)

Chitalo Mahabidyalaya, Jajpur, Odisha
05/2024

Secondary School Examination (10th)

Mahabir Nodal High School, Jajpur, Odisha
05/2022

CERTIFICATIONS & ACHIEVEMENTS

Certified in Advanced Drone Technology

I-STEM & NIELIT
2025

Smart India Hackathon (SIH)

Participant Digital Health Record Management System 2025

Published Technical Paper

Digital Health Record Systems
2025

PROFESSIONAL SUMMARY

ECE undergraduate specializing in RTL Design, Functional Verification, and ASIC Physical Implementation. Skilled in Verilog/SystemVerilog, OpenLane, logic synthesis, and static timing analysis, with practical experience in digital design and physical-design workflows.

TECHNICAL SKILLS

Hardware Description Languages

Verilog HDL • SystemVerilog (Basics)

VLSI & Physical Design

RTL Design • Logic Synthesis • Floorplanning • Power Distribution Network (PDN) • Placement & Routing (P&R;) • Clock Tree Synthesis (CTS) • Static Timing Analysis (STA) • Parasitics Extraction (SPEF) • Physical Verification (DRC/LVS)

EDA Tools & Toolchains

OpenLane Flow • OpenROAD • Yosys • Magic VLSI • KLayout • Xilinx Vivado • Netgen

Process Nodes & PDKs

SkyWater 130nm (sky130A) • sky130_fd_sc_hd Standard Cell Library

Platforms & Environments

Linux / Ubuntu • WSL2 • Docker Desktop • Windows 11

TRAINING / COURSES

Semiconductor Technology Course

IIT Roorkee
06/2026 – 08/2026

- Completed specialized coursework on semiconductor packaging techniques and low-power CMOS technology.

VLSI Design Intern

CTTC, Bhubaneswar
06/2026 – 07/2026

- Developed and simulated 4-bit ALU and basic CPU datapath blocks in Verilog HDL.
- Analyzed behavioral simulation waveforms and digital combinational logic optimization.

Python Programming Course

Pythonsoft LLP, Bhubaneswar
11/2025 – 04/2026

- Built foundational concepts in data structures, algorithms, and automated scripting.

CORE PHYSICAL DESIGN

RTL Design • Logic Synthesis • Floorplanning • PDN • Placement & Routing • CTS • STA • SPEF • DRC/LVS

FIND ME ONLINE

GitHub

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